Radioactivity

Saurabh Sandilya

saurabh@phy.iith.ac.in



भारतीय सांकेतिक विज्ञान संस्थ हैदराबाद
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

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Radioactivity

- Radioactivity is the spontaneous emission of particles/EM radiation from unstable atomic nuclei.
- This process, also known as radioactive decay, transforms an unstable parent nucleus into a more stable daughter nucleus.
- The emitted radiation can be in the form of particles (like alpha or beta particles) or electromagnetic energy (like gamma rays).
- Radioactive decay is a random process governed by the laws of probability.
- The rate of decay is unique to each radioactive isotope and is measured by its half-life.

Radioactivity: The Accidental Discovery



- 1883: Henri Becquerel started studying fluorescence and phosphorescence.
- 1896: Becquerel started investigating the connection between phosphorescence and X-rays discovered in the same year by Roentgen.
- He placed a uranium salt on a photographic plate wrapped in black paper, hoping sunlight would induce X-rays.
- Due to cloudy weather, the setup was stored in a dark drawer.
- Upon developing the plate, he found the uranium crystal's silhouette clearly exposed the film.

Source: https://www.aps.org/apsnews/2008/02/becquerel-discovers-radioactivity

Radioactivity: The Accidental Discovery

- Becquerel concluded: The uranium salt emitted radiation (which he called 'uranic rays') independently of sunlight or phosphorescence.

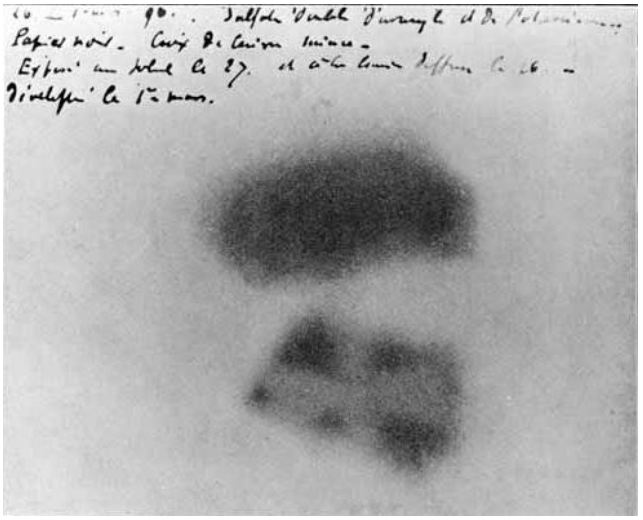
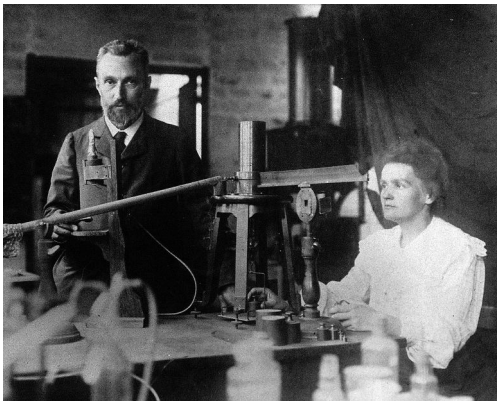


Image of Becquerel's photographic plate, which has been fogged by exposure to radiation from a uranium salt. The shadow of a metal Maltese Cross placed between the plate and the uranium salt is clearly visible.

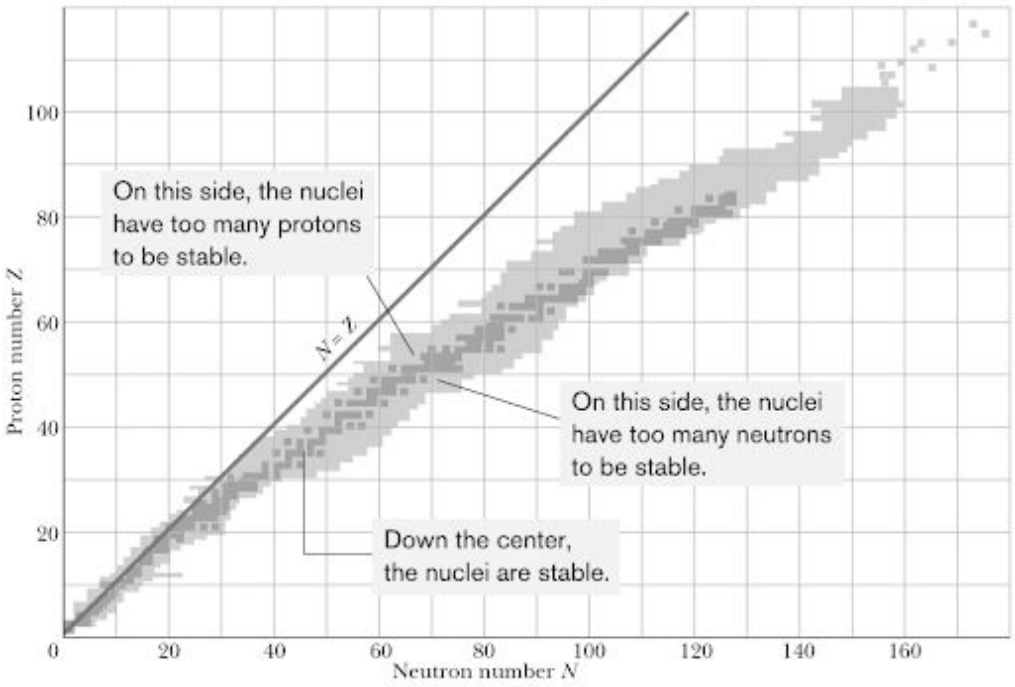
Source: Wikipedia

Radioactivity: The Curies' groundbreaking research



- Madam Curie decided to do her thesis on radiation, recently discovered in uranium by Becquerel.
- Used a sensitive electrometer to measure radiation, establishing that its intensity was proportional to the amount of uranium.
- Coined the term 'radioactivity' to describe this spontaneous emission of radiation.
- Observed that the ore pitchblende was far more radioactive than pure uranium, correctly deducing the presence of undiscovered radioactive elements.
- Through painstaking chemical work with Pierre, she isolated two new, highly radioactive elements: Polonium (named after Poland) and Radium.

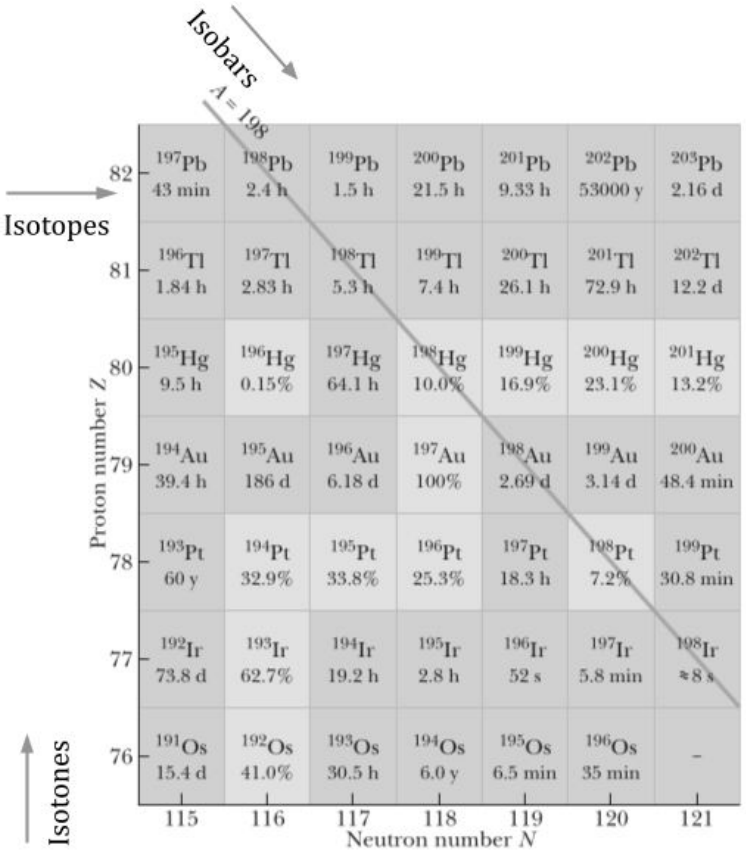
N *vs.* Z plot and nuclear stability



Reference: Halliday/Resnick/Walker

- Low-mass stable nuclides typically possess an equal number of neutrons and protons.
- More massive nuclides exhibit an increasing surplus of neutrons.

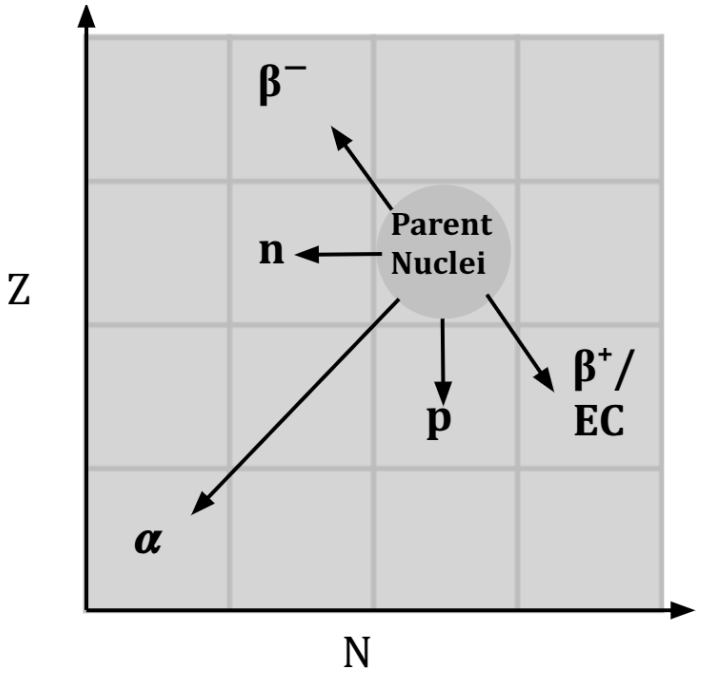
N *vs.* Z zoomed plot



- Isotopes: Same atomic number (Z); different mass numbers (A); and different numbers of neutrons (N).
- Isotones: Different atomic number (Z); different mass numbers (A); and same numbers of neutrons (N).
- Isobars: Different atomic number (Z); same mass numbers (A); and different numbers of neutrons (N).

Radioactive displacement law

- The Radioactive Displacement Law, also known as the Soddy-Fajans-Russell Law, describes how a parent nuclide changes into a daughter nuclide during radioactive decay.



Radioactive displacement law

- **Alpha (α) Decay:** A of the daughter nucleus reduces by 4, and Z reduces by 2 (N also reduces by 2).

$$U_{92}^{238} \rightarrow Th_{90}^{234} + He_{2}^{4}$$

- **Beta (β^-) Decay:** Z increases by one (N decreases by one), A remains unchanged.

$$n \rightarrow p + e^- + \bar{\nu}_e$$

$$C_{6}^{14} \rightarrow N_{7}^{14} + e^- + \bar{\nu}_e$$

- **Beta (β^+) Decay & Electron Capture:** Z decreases by one (N increases by one), A remains unchanged.

(β^+ decay)

$$p \rightarrow n + e^+ + \nu_e$$

$$F_{9}^{18} \rightarrow O_{8}^{18} + e^+ + \nu_e$$

(electron capture)

$$p + e^- \rightarrow n + \nu_e$$

$$K_{19}^{40} + e^- \rightarrow Ar_{18}^{40} + \nu_e$$

The Radioactivity decay law

- Lifetimes of unstable nuclei are statistical in nature.
- If we observe a collection of some unstable nuclei, they don't decay at the same time.
- Consider the disintegration of a 'nuclei $P \rightarrow$ nuclie Q':
  - number of nuclei P decreases with time.
  - number of nuclei Q increases with time.
- Let N be the (large) number of nuclei P in a sample.
- It is found experimentally that the sample activity (= rate of change of 'N' with time = $\frac{dN}{dt}$) is proportional to the number of nuclei (N) in the sample.

$$\frac{dN}{dt} \propto N$$

$$-\frac{dN}{dt} = \lambda N$$

- λN = Activity of the sample.
- λ = Disintegration constant.

- This means the more nuclei you have, the more decays occur per second.

The Radioactivity decay law

In a sample of large N nuclei, the mean number of nuclei decaying in a time interval dt would be

$$\Rightarrow dN = -\lambda N dt$$

Integrating the above equation

$$\Rightarrow \int_{N(0)}^{N(t)} \frac{dN}{N} = -\int_0^t \lambda dt$$

$$\Rightarrow [\ln N]_{N(0)}^{N(t)} = -[\lambda t]_0^t$$

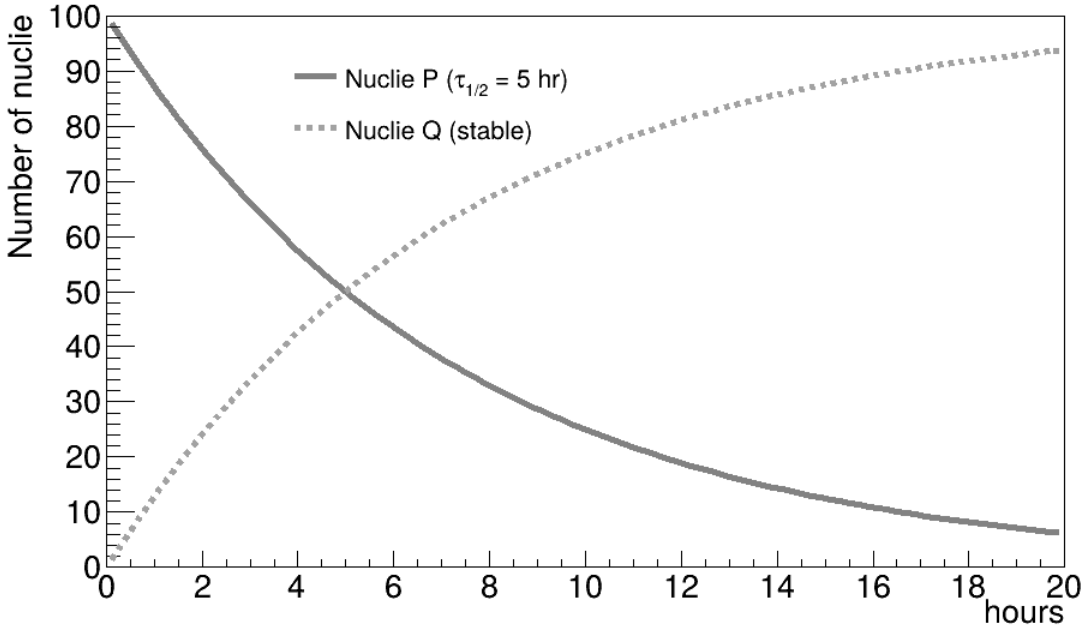
$$\Rightarrow \ln N(t) - \ln N(0) = -\lambda t$$

$$\Rightarrow \ln\left[\frac{N(t)}{N(0)}\right] = -\lambda t$$

$$\Rightarrow \boldsymbol{N(t) = N(0).exp(-\lambda t)}$$

The Radioactivity decay law

- Number of parent (decaying) nuclide (P) sample at t: $N(t) = N(0).exp(-\lambda t)$.
- Number of daughter nuclide (Q) sample at t: $N(t) = N(0).[1 - exp(-\lambda t)]$.



Probability Density for the decay

- Number of nuclei (P) in time $(t + dt)$:

$$N(t + dt) = N(0) \exp[-\lambda(t + dt)]$$
$$N(t + dt) = N(0).\exp(-\lambda t).\exp(-\lambda dt)$$
$$N(t + dt) \simeq N(0).\exp(-\lambda t).(1 - \lambda dt)$$

- So, the number of nuclei decayed between times (t) and $(t + dt)$:

$$N(t) - N(t + dt) = N(0).\exp(-\lambda t) - N(0).\exp(-\lambda t).(1 - \lambda dt)$$
$$N(t) - N(t + dt) = N(0).\exp(-\lambda t).\lambda dt$$

- Probability that an individual nuclie (selected at random) from the initial (at $t = 0$) sample will decay between t and $t + dt$:

$$P(t)dt = \frac{N(0).\exp(-\lambda t).\lambda dt}{N(0)}$$
$$P(t)dt = \exp(-\lambda t).\lambda dt$$

- $P(t)$: Probability that an individual nuclie (selected at random) from the initial (at $t = 0$) sample will decay at time t.

Average lifetime

- To calculate the average/mean lifetime :

$$\tau = \langle t \rangle = \int_0^\infty t.P(t)dt$$
$$\tau = \int_0^\infty t.exp(-\lambda t)\lambda dt = \lambda \int_0^\infty t.exp(-\lambda t)dt$$

- (integration by parts)

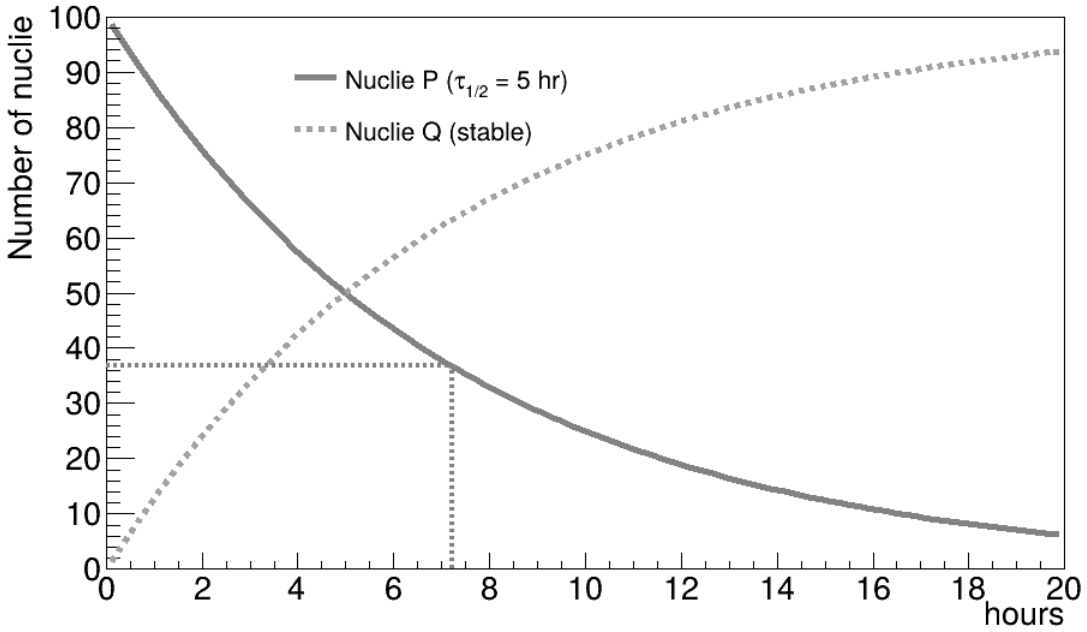
$$\tau = \cancel{\left[-t.e^{-\lambda t}\right]_0^\infty} \to 0 + \int_0^\infty e^{(-\lambda t)}dt$$
$$\tau = \int_0^\infty e^{(-\lambda t)}dt = \left[-\frac{1}{\lambda}.e^{(-\lambda t)}\right]_0^\infty$$
$$\tau = 0 + \frac{1}{\lambda}.e^{(-\lambda 0)} = \frac{1}{\lambda}$$

- Mean/average lifetime: $\tau = \frac{1}{\lambda}$

Sample size after mean lifetime

- What is the sample size after 'mean life time' : $t = 1/\lambda$?

$$N(\tau) = N(0).exp(-\lambda\tau) = N(0).exp(-\lambda.\frac{1}{\lambda}) = \frac{N(0)}{e} = \frac{N(0)}{2.718}$$



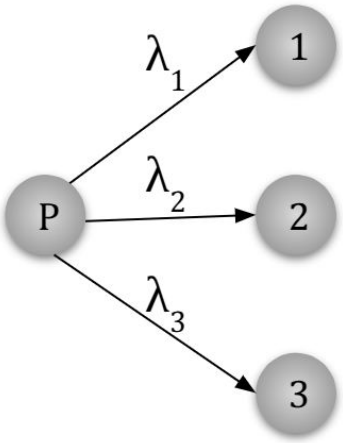
Half-Life

- Half-Life: Large sample of radioactive nuclei is reduced to half ($\frac{1}{2}$) of the number at t=0

$$N(\tau_{\frac{1}{2}}) = N(0).exp(-\lambda\tau_{\frac{1}{2}})$$
$$\frac{N(0)}{2} = N(0).exp(-\lambda\tau_{\frac{1}{2}})$$
$$exp(\lambda\tau_{\frac{1}{2}}) = 2$$
$$\lambda\tau_{\frac{1}{2}} = ln(2)$$
$$\tau_{\frac{1}{2}} = ln(2)/\lambda$$
$$\tau_{\frac{1}{2}} = \tau.ln(2)$$

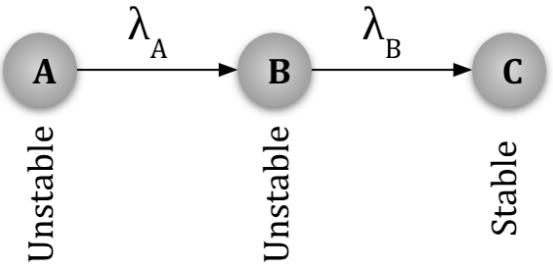
Multiple Decay Modes

- Let us consider three competing branches in the decaying parent nuclei P to daughter nuclei 1 , 2 and 3 .
- The decay rates λ_1, λ_2 and λ_3 define probability per unit time for disintegration, P $\rightarrow$ 1 , P $\rightarrow$ 2 and, P $\rightarrow$ 3 , respectively.



- The total probability for disintegration (for P) is: $\lambda = \lambda_1 + \lambda_2 + \lambda_3$
- Relative probability for each branch decay, called the branching ratio, is the ratio of the respective decay rate to the total decay rate: $\mathcal{B}[\text{P} \rightarrow 1] = \lambda_1/\lambda$, $\mathcal{B}[\text{P} \rightarrow 2] = \lambda_2/\lambda$, $\mathcal{B}[\text{P} \rightarrow 3] = \lambda_3/\lambda$.
- And, also: $\frac{1}{\tau} = \frac{1}{\tau_1} + \frac{1}{\tau_2} + \frac{1}{\tau_3}$ (similar relation with half-life)

Successive Disintegration



- The disintegration rates can be written as:
  - A decays at the rate $\lambda_A N_A$:

$$\frac{dN_A}{dt} = -\lambda_A N_A \tag{1}$$

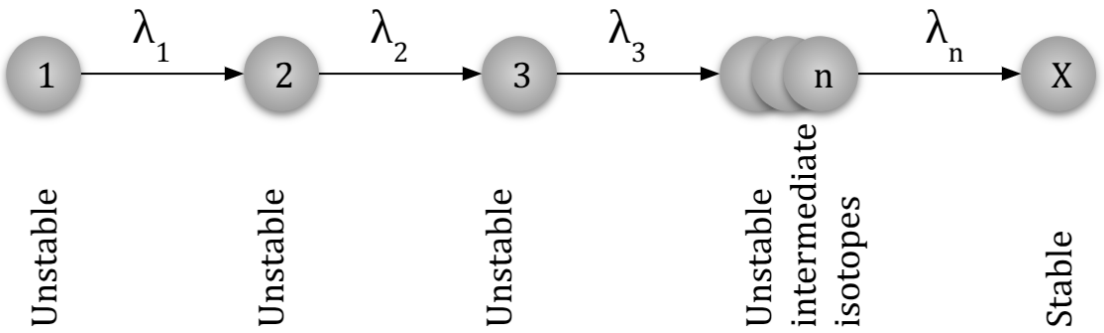
  - B is produced at the rate $\lambda_A N_A$, and decays at the rate $\lambda_B N_B$:

$$\frac{dN_B}{dt} = \lambda_A N_A - \lambda_B N_B \tag{2}$$

  - C is produced at the rate $\lambda_B N_B$, and doesn't decays

$$\frac{dN_C}{dt} = \lambda_B N_B \tag{3}$$

Successive Disintegration



$$\frac{dN_1}{dt} = -\lambda_1 N_1$$
$$\frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2$$
$$\frac{dN_3}{dt} = \lambda_2 N_2 - \lambda_3 N_3$$
$$\frac{dN_n}{dt} = \lambda_{n-1} N_{n-1} - \lambda_n N_n$$

- **Bateman Equation:** A mathematical tool for decay chain analysis.
- It models how nuclides in a chain evolve over time.
- **History:** Conceived by Rutherford, solved by Bateman.

Radioactive decay series

- A heavy radionuclide may be too far from the valley of stability to reach a stable nucleus in a single decay.
- The daughter produced in one decay can itself be radioactive, giving rise to a **decay chain**
- A radioactive series is a sequence of nuclear decays in which an unstable parent transforms through successive daughter nuclei until a stable nuclide is reached.
- Four chains are classified according to the mass-number remainder modulo 4.

Series	Mass-number class	Representative parent
Thorium	$A = 4n$	^{232}Th
Neptunium	$A = 4n + 1$	^{237}Np
Uranium	$A = 4n + 2$	^{238}U
Actinium	$A = 4n + 3$	^{235}U

Why exactly four series?

- α decay: ${}^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\mathrm{He}$
  - $A \to A-4$
  - $Z \to Z-2$
- β^- decay : ${}^{A}_{Z}X \longrightarrow {}^{A}_{Z+1}Y + e^- + \bar{\nu}_e$
  β^+ decay : ${}^{A}_{Z}X \longrightarrow {}^{A}_{Z-1}Y + e^+ + \nu_e$
  - A unchanged
  - $Z \to Z+1$ (for β^-), or $Z \to Z-1$ (for β^+)
- Only α decay changes the mass-number class A (mod 4). Mass number splits into four modular families:
  $A = 4n + k \quad (k \in \{0, 1, 2, 3\})$
- Decay Stoichiometry Equations:

$$N_\alpha = \frac{A_{\text{parent}} - A_{\text{final}}}{4}, \quad N_\beta = 2N_\alpha - (Z_{\text{parent}} - Z_{\text{final}})$$

Primordial Decay Chains ($4n, 4n+2, 4n+3$)

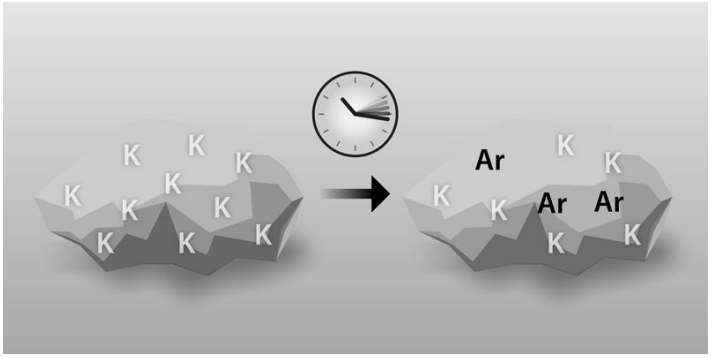
- **Thorium Series ($4n$):**
  - Initiates at ${}^{232}\mathrm{Th}_{90}$ ($t_{1/2} \approx 14.05$ Gyr) $\to$ terminates at ${}^{208}\mathrm{Pb}_{82}$ ($6\alpha + 4\beta^-$)
  - Intermediary gas: Thoron (${}^{220}\mathrm{Rn}_{86}$, $t_{1/2} = 55.6$ s)
- **Uranium Series ($4n+2$):**
  - Initiates at ${}^{238}\mathrm{U}_{92}$ ($t_{1/2} \approx 4.47$ Gyr) $\to$ terminates at ${}^{206}\mathrm{Pb}_{82}$ ($8\alpha + 6\beta^-$)
  - Emits Radon (${}^{222}\mathrm{Rn}_{86}$, $t_{1/2} = 3.82$ d), a major terrestrial hazard
- **Actinium Series ($4n+3$):**
  - Initiates at ${}^{235}\mathrm{U}_{92}$ ($t_{1/2} \approx 0.704$ Gyr) $\to$ terminates at ${}^{207}\mathrm{Pb}_{82}$ ($7\alpha + 4\beta^-$)
  - Isotopic abundance of ${}^{235}\mathrm{U}$ reduced from $\approx 24\%$ (Earth formation) to $\approx 0.72\%$ today

Extinct $4n+1$ Series & Cosmochemical Significance

- **Cosmochemical Survival Condition:**
  - Primordial nuclides require $t_{1/2} \gtrsim t_{\text{Earth}} \approx 4.54 \times 10^9$ yr
- **Neptunium Chain Dynamics:**
  - Longest-lived member: $^{237}\text{Np}_{93}$ ($t_{1/2} = 2.14 \times 10^6$ yr)
  - Depletion factor over Earth's age: $\exp(-\lambda_{\text{Np}} t_{\text{Earth}}) \approx 10^{-652}$
  - Extinct naturally; recreated in reactors via $^{238}\text{U}(n,2n)^{237}\text{U} \xrightarrow{\beta^-} {}^{237}\text{Np}$
- **Terminal Nuclide Dynamics:**
  - Ends at $^{209}\text{Bi}_{83}$ via $7\alpha + 4\beta^-$ decays
  - *Note:* ^{209}Bi alpha decays to ^{205}Tl with $t_{1/2} = (1.9 \pm 0.2) \times 10^{19}$ yr

Radioactive Dating: $K - Ar$ Dating

- Parent Isotope: K^{40} is present in the molten rock (magma).
- Decay Process: A small fraction of the K^{40} in the rock decays $\tau_{1/2} = 1.25 \times 10^9$ y into Ar^{40}, a stable, inert noble gas.



Phys. Rev. Lett. 131, 052503 (2023)

- When magma cools and solidifies, forming solid rock, the gaseous Ar^{40} escapes. This event effectively 'resets the clock.'
- From this point on, any produced Ar^{40}, by the decay of the trapped K^{40} is held within the rock's crystal structure.

$$t = \frac{\tau_{1/2}.ln(1+N_{Ar}/N_K)}{ln(2)}$$

- Dating rocks from Earth, the Moon, and meteorites shows a consistent maximum age of about 4.5 billion years.

Derivation: $K - Ar$ Dating

- A sample solidifies with $N_K(0)$ atoms of radioactive ^{40}K. At a later time t: $N_K(t) = N_K(0)\, e^{-\lambda t}$, where $\lambda = \frac{\ln 2}{\tau_{1/2}}$.
- Assume every ^{40}K nucleus that decays produces a stable ^{40}Ar atom. Then $N_{Ar}(t) = N_K(0) - N_K(t)$.

$$\Rightarrow N_{Ar}(t) = N_K(t)\, e^{\lambda t} - N_K(t) = N_K(t)\left(e^{\lambda t} - 1\right)$$

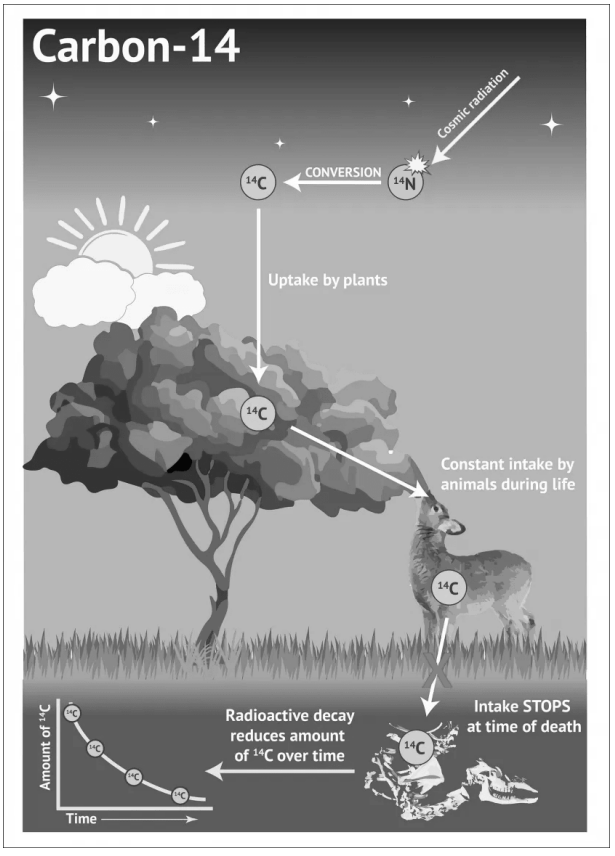
$$\Rightarrow \frac{N_{Ar}}{N_K} = e^{\lambda t} - 1 \quad \Longrightarrow \quad e^{\lambda t} = 1 + \frac{N_{Ar}}{N_K}.$$

$$\Rightarrow \lambda t = \ln\left(1 + \frac{N_{Ar}}{N_K}\right) \quad \Longrightarrow \quad t = \frac{1}{\lambda} \ln\left(1 + \frac{N_{Ar}}{N_K}\right).$$

$$\boxed{t = \frac{\tau_{1/2}\, \ln(1 + N_{Ar}/N_K)}{\ln 2}}$$

Radioactive Dating: $C - 14$ Dating

- Radioactive C^{14} has a $\tau_{1/2}$ of 5,730 years.
- Cosmic rays bombard N^{14} in the upper atmosphere to produce C^{14} continuously.
- Clock starts when an organism dies, and exchange of stops C^{14}. The amount of C^{14} trapped within it begins to decay.
- By measuring the remaining ratio $C^{14} : C^{12}$, we can determine the time elapsed since the organism died.
- Ideal for dating organic materials up to $\sim 50,000$ years old.



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The End